AI Research Assistant Platform

An n8n automation that collects, filters, summarises, cites and delivers research papers

n8n Capstone Project · Assignment 8 · IIT Jammu AI/LLM Summer School 2026 · Anshika Rana



Register
workflow 01

Collect
workflow 02

Summarise
workflow 03

Cite & archive
workflow 04

Digest
workflow 05

Audit
workflow 06

Keeping up with research is slow, manual and repeated

The same searches, every week

Queries are re-typed on arXiv, Semantic Scholar and publisher sites

Too much to read

Keyword search returns many papers that only mention the term

Inconsistent notes

Summaries miss the method, datasets or limitations

Messy references

Hand-copied citations in mixed formats, with duplicates

New work is missed

Nobody is notified when a matching paper appears

Stakeholders

Researchers & students

relevant papers, trustworthy summaries, ready citations

Faculty supervisors

a quick view of a topic; consistent references

Lab admin

no secrets in code, audit trail, alerts, controlled LLM cost

Library / knowledge mgmt

a searchable repository with stable IDs and PDFs

Each objective in the brief maps to one workflow

Workflow 01

Register topics

Web form + REST API, validation, duplicate check

Workflow 02

Collect papers daily

arXiv + Semantic Scholar daily, merged & de-duplicated, AI relevance ≥ 0.70

Workflow 03

AI-powered summaries

10-field structured note per paper with GPT-5-mini

Workflow 04

Organise references

APA + BibTeX in code, PDF archive, search API

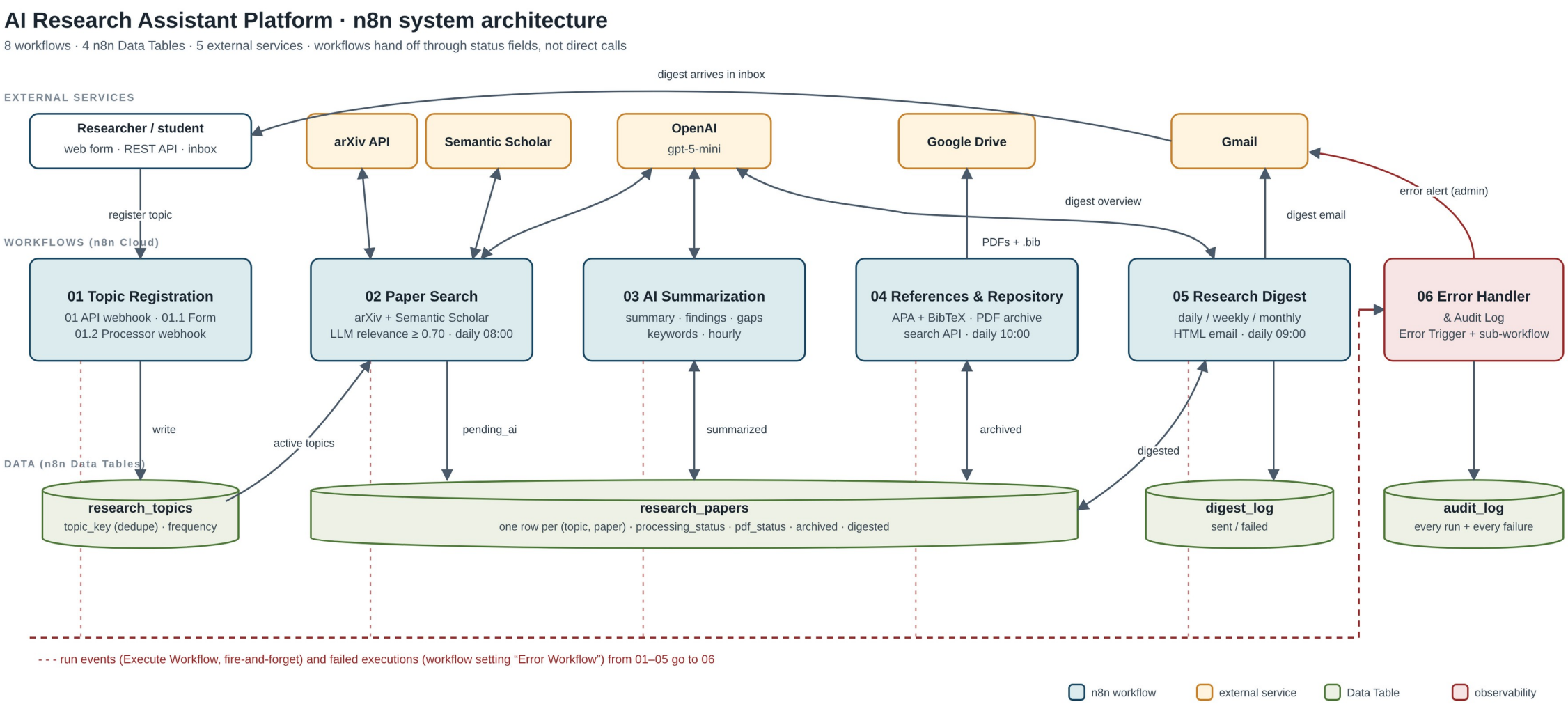
Workflow 05

Track new publications

Personal digest: daily, weekly or monthly, with an AI overview

Workflow 06 · Error Handler & Audit Log covers the brief's advanced features: error handling, retries, logging and an audit trail for every run and every failure.

System architecture: 8 workflows, 4 Data Tables, 5 services



Workflows hand off work through status fields, not direct calls



Each workflow reads rows in one state from the Data Tables, does its job and writes the next state.

Independent stages

Each workflow has its own schedule and can be re-run or debugged alone.

Failures stay contained

A failed stage never corrupts another stage. Its rows simply wait in their current state.

Status is visible

A paper's row shows exactly where it is in the pipeline, which makes the audit easy.

Registration: validated, de-duplicated, audited

POST /webhook/research/topic

```
{
  "researcher": "Anshika Rana",
  "email": "you@example.com",
  "topic": "Retrieval Augmented Generation",
  "keywords": "RAG, retrieval, LLM",
  "frequency": "weekly"
}
```

Also available as an n8n web form (01.1), which posts to a processor webhook (01.2) and shows one of three result pages.

200 — **Registered**

topic_id TOPIC-<ms>-<rand>; keywords lower-cased and de-duplicated

400 — **Invalid**

All problems listed at once: name, email, topic length, keywords, frequency

409 — **Duplicate**

topic_key = email|topic ignores case and spaces, so a researcher can't register the same topic twice

Collection: two sources, one clean list, an AI relevance gate

1. Daily 08:00: one query per active topic
2. arXiv + Semantic Scholar in parallel: rate-limited, 30 s timeout, 3 retries
3. Normalise both formats; merge duplicates by arXiv ID → DOI → title + year
4. Skip papers already stored for this topic, before any LLM call
5. GPT-5-mini scores relevance 0–1 with a reason; output validated
6. Store papers scoring ≥ 0.70 as pending_ai; log the rest

37

candidates after de-duplication (3 topics)

36

stored as relevant (≥ 0.70)

1

filtered out by the AI gate

0

AI failures in the live run

Live run, 7 Oct 2026 (execution #86, 66 s). Stored papers scored 0.92–0.98.

Summarisation: every paper becomes a structured research note

10 fields per paper (strict JSON, validated)

`summary`	`key_findings`
`methodology`	`datasets`
`models`	`limitations`
`research_gap`	`research_significance`
`extracted_keywords`	`research_category`

The prompt forbids anything not in the abstract. Invalid output → ai_failed, never a broken run.

Real output · FAIR-RAG (arXiv 2510.22344)

Summary An agentic RAG framework that decomposes a query into required findings, audits the evidence for gaps and issues targeted sub-queries until the evidence is sufficient.

Key finding F1 0.453 on HotpotQA, +8.3 points over the strongest iterative baseline.

Datasets HotpotQA · 2WikiMultiHopQA · MusiQue

Research gap A structured, evidence-driven process that explicitly finds and fills information gaps for multi-hop reasoning.

36 / 36 papers summarised · 0 AI failures

References, PDF archive and a searchable repository

Citations are built in code, not by the LLM

Asl, M. A., Asgari-Bidhendi, M., & Minaei-Bidgoli, B. (2025). FAIR-RAG: Faithful Adaptive Iterative Refinement for Retrieval-Augmented Generation. arXiv preprint arXiv:2510.22344.

```
@misc{asl2025fair,
  title = {FAIR-RAG: Faithful Adaptive ...},
  author = {Mohammad Aghajani Asl and ...},
  year = {2025}, eprint = {2510.22344},
  archivePrefix = {arXiv} }
```

PDF outcome per paper

`stored` `drive_upload_failed` `download_failed` `no_link`

Search API

```
GET /webhook/research/papers/search
?q=faithfulness attribution
```

11 VISA: RAG with Visual Source Attribution

7 RAG in Quranic Studies: 13 open-source LLMs

6 Overview of the TREC 2025 RAG Track

Ranking: title ×3, keywords ×2, summary and abstract ×1. ?paper_id= returns related papers.

The digest: an AI-written briefing, not just a list of links

Research digest: Retrieval Augmented Generation – 10 new paper(s)

"These papers show RAG moving from proof-of-concept demos toward engineering-first, trust-oriented systems that prioritize faithfulness, attribution, and domain adaptivity."

Emerging trends

- Faithfulness and attribution-first RAG
- Maturing benchmarks and evaluation (shared tracks, multi-dimensional metrics)
- Modular, agentic and multi-agent architectures

Read first: FAIR-RAG · Engineering the RAG Stack · TREC 2025 RAG Track overview

Then one card per paper: summary, key findings, PDF link, APA citation.



On the researcher's schedule

daily · weekly on Mondays · monthly on the 1st (09:00 IST)



AI overview with a fallback

If the LLM call fails, the digest still goes out with a plain list



Safe delivery

A failed send is logged; papers stay undigested and the next run retries

Built to fail safely, and tested on every failure path

Situation	What the system does
Invalid or duplicate topic	400 with every error, or 409. Nothing is written
arXiv / Semantic Scholar down or rate-limited	Timeout, 3 retries with back-off, then 0 results from that source
LLM fails or returns bad JSON	Validated: score 0 (02), ai_failed (03), plain-list digest (05)
Broken PDF link or Drive failure	pdf_status records the outcome; the run continues
Email send fails	digest_log = failed; papers retried in the next run
Any unhandled node error	Error Workflow 06 → audit_log + admin alert

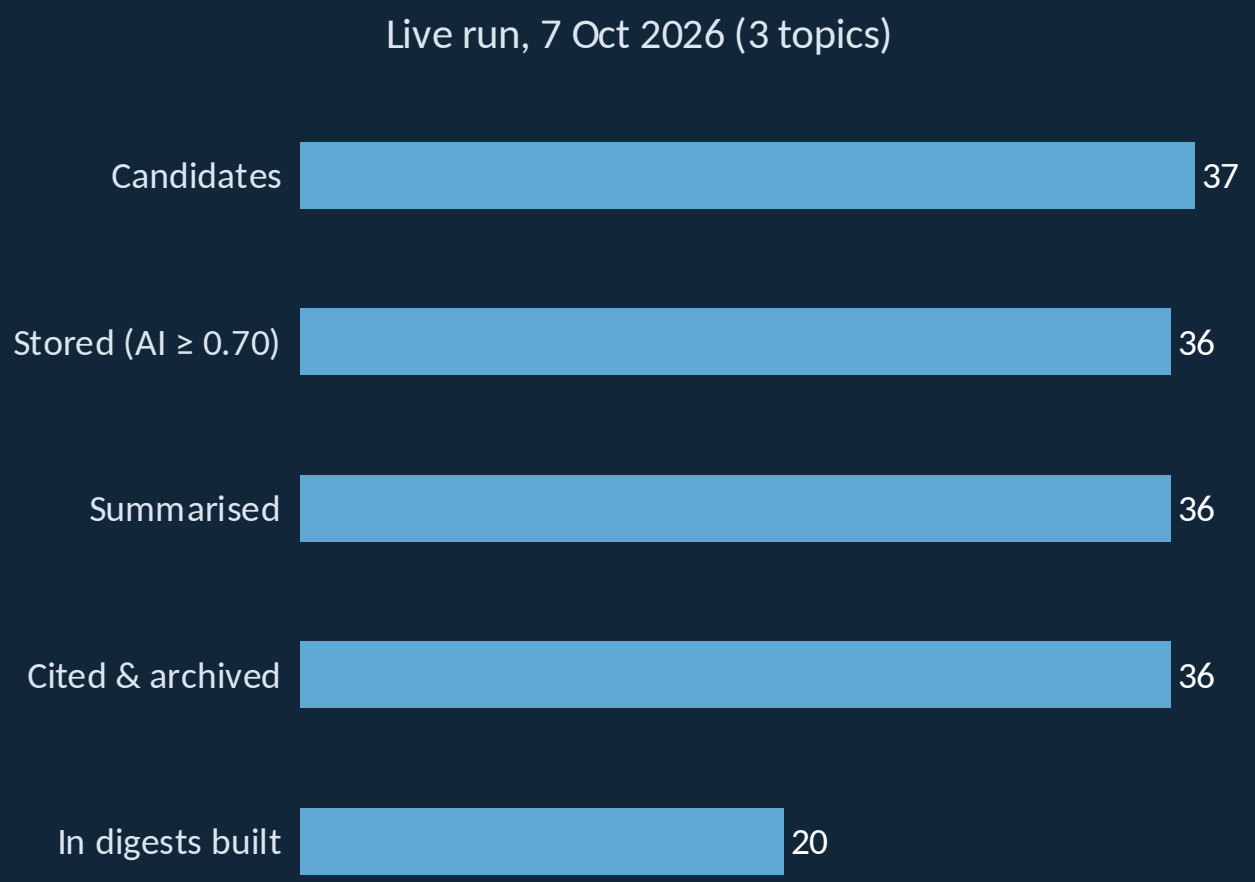
Bug: with 25 PDFs failing at once, items lost their paired link. Fix: an "Attach Reference" node carries each paper's data along with its PDF.

14 scenario tests passed (T1–T14)

7 live pipeline steps on real data

1 real bug found and fixed

Results and next steps



What I learned

- Use AI for judgement, code for facts: relevance and summaries by LLM; citations, de-duplication and search in code
- Treat LLM output as untrusted input and give every failure an explicit branch
- Loose coupling through Data Tables makes each stage easy to test and re-run

Next steps

- Connect Gmail, Google Drive and Semantic Scholar on the live instance
- Full-text PDF summaries and a citation-based "related papers" graph
- A human approval step before a digest is sent